

Lab 12

Two Port Network

Name: Syed Muhammad Asad Ali	Student ID: sa09475
------------------------------	---------------------

12.1 Objective

To represent a two-port network into its equivalent impedance, transmission and admittance parameter models.

12.2 Introduction

A pair of terminals from any linear electrical network is called **port**. A single-port network has two terminals i.e., one port. The voltage between the terminals is called port voltage and the current into the terminal is called port current. Similarly, a two-port network has 4 terminals and two ports as represented in Figure 12.1.

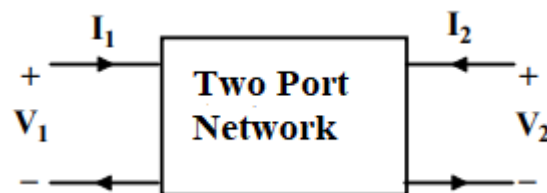


Figure 12.1: Two-Port Network Representation

For analyzing purposes, any electrical network systems such as amplifiers, transistors, filters, matching circuits, transformers, transmission cable etc. can be modeled as a two-port network; input port at left side and output port at right side. A complex circuit when represented by a two-port network becomes a black box. This isolated two-port network simplifies network analysis by representing complex connections and component values in a circuit into their equivalent parameters. To transform a linear circuit into a two-port network, the following conditions must be met.

1. No energy will be stored in two-port network
2. No independent source exists in two-port network
3. The same current must enter and leave a port
4. External connections must be made to either the input or output port i.e., no connections are allowed between the ports

Equivalent Parameters: A two-port network can be modelled using equivalent parameters which are listed below:

1. Impedance z-parameter
2. Admittance y-parameter
3. Hybrid h-parameter



4. Transmission t or ABCD parameter
5. Inverse hybrid g-parameters

The equations used to describe the above parameters are given in Table 12.1.

Table 12.1

Impedance z-Parameters		
$V_1 = z_{11}I_1 + z_{12}I_2$	$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ -I_2 \end{bmatrix}$	$z_{11} = \left. \frac{V_1}{I_1} \right _{I_2=0}$ (Open circuit input impedance)
$V_2 = z_{21}I_1 + z_{22}I_2$		$z_{12} = \left. \frac{V_1}{I_2} \right _{I_1=0}$ (Transfer impedance from port 1 to 2) $z_{21} = \left. \frac{V_2}{I_1} \right _{I_2=0}$ (Transfer impedance from port 2 to 1) $z_{22} = \left. \frac{V_2}{I_2} \right _{I_1=0}$ (Open circuit output impedance)
Admittance y-Parameters		
$I_1 = y_{11}V_1 + y_{12}V_2$	$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$	$y_{11} = \left. \frac{I_1}{V_1} \right _{V_2=0}$ (Short circuit input admittance)
$I_2 = y_{21}V_1 + y_{22}V_2$		$y_{12} = \left. \frac{I_1}{V_2} \right _{V_1=0}$ (Transfer admittance from port 1 to 2) $y_{21} = \left. \frac{I_2}{V_1} \right _{V_2=0}$ (Transfer admittance from port 2 to 1) $y_{22} = \left. \frac{I_2}{V_2} \right _{V_1=0}$ (Output short circuit admittance)
Hybrid h-Parameters		
$V_1 = h_{11}I_1 + h_{12}V_2$	$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$	$h_{11} = \left. \frac{V_1}{I_1} \right _{V_2=0}$ (Short circuit input impedance)
$I_2 = h_{21}I_1 + h_{22}V_2$		$h_{12} = \left. \frac{V_1}{V_2} \right _{I_1=0}$ (Open circuit reverse voltage gain) $h_{21} = \left. \frac{I_2}{I_1} \right _{V_2=0}$ (Short circuit forward current gain) $h_{22} = \left. \frac{I_2}{V_2} \right _{I_1=0}$ (Open circuit output admittance)
Transmission t-Parameters or ABCD Parameters		

$V_1 = AV_2 - BI_2$	$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ I_2 \end{bmatrix}$	$A = \left. \frac{V_1}{V_2} \right _{I_2=0}$ (Open circuit voltage ratio)
$I_1 = CV_2 - DI_2$		$B = \left. \frac{V_1}{-I_2} \right _{V_2=0}$ (Short circuit transfer impedance)
		$C = \left. \frac{I_1}{V_2} \right _{I_2=0}$ (Open circuit transfer admittance)
		$D = \left. \frac{I_1}{-I_2} \right _{V_2=0}$ (Short circuit current ratio)
Inverse Hybrid g-Parameters		
$I_1 = y_{11}V_1 + y_{12}I_2$	$\begin{bmatrix} I_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ I_2 \end{bmatrix}$	$g_{11} = \left. \frac{I_1}{V_1} \right _{I_2=0}$ (Open circuit input admittance)
$V_2 = y_{21}V_1 + y_{22}I_2$		$g_{12} = \left. \frac{I_1}{I_2} \right _{V_1=0}$ (Short circuit reverse current gain)
		$g_{21} = \left. \frac{V_2}{V_1} \right _{I_2=0}$ (Open circuit forward voltage gain)
		$g_{22} = \left. \frac{V_2}{I_2} \right _{V_1=0}$ (Short circuit output impedance)

Procedure:

The circuit given in Figure 12.2. can be modelled as a two-port network. In the following tasks, impedance and transmission parameters will be determined and verified for this circuit.

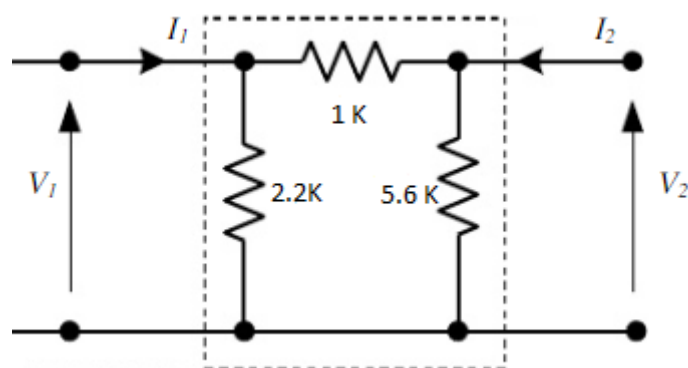
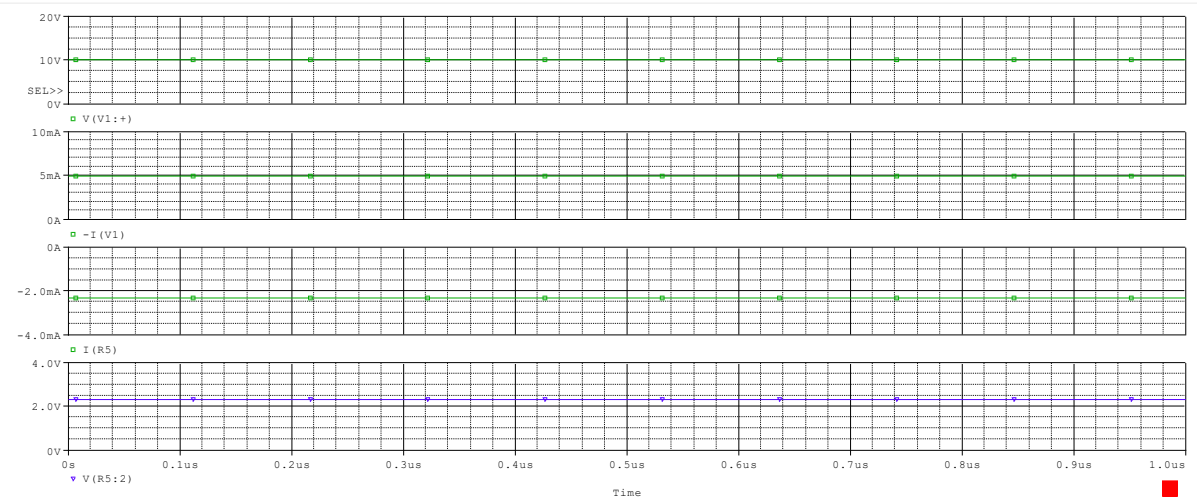
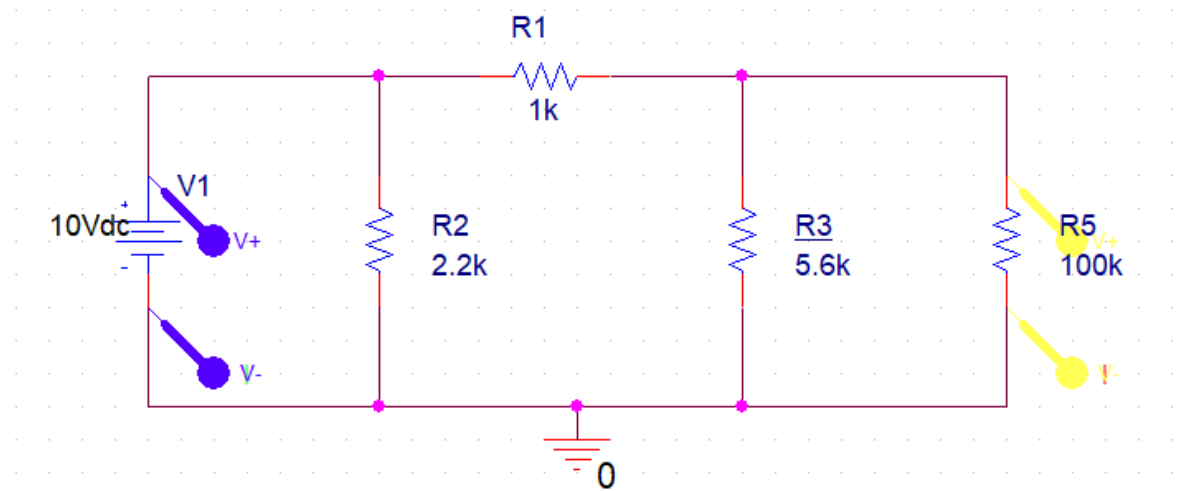


Figure 12.2

Task 1: To find impedance-parameters

1. Apply DC supply of 10 V at input port in PSPICE and leave output port open-circuited i.e. $I_2 = 0$ (place a very high value resistance at output to make $I_2=0$). Now measure input and output voltages and currents.

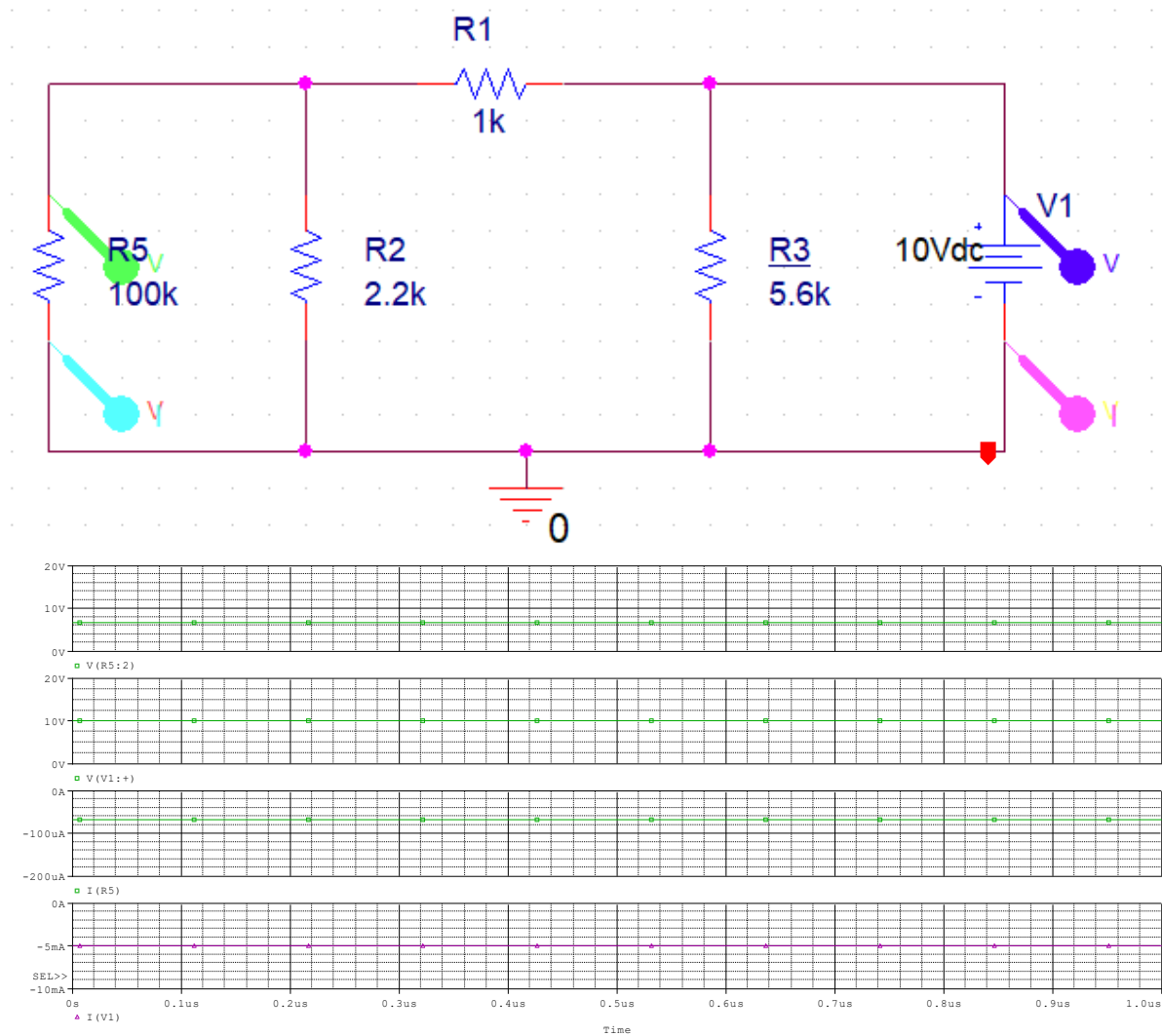


$$V_1 = 10V \quad V_2 = 8.4135V \quad I_1 = 6.132mA \quad I_2 = 84.135uA.$$

The open circuit output refers to the condition for z_{11} and z_{21} . Therefore, using above values and formulae given in Table 12.1, calculate z_{11} and z_{21} .

$$z_{11} = 1.6308k \text{ ohms} \quad z_{21} = 1.372k \text{ ohms}$$

2. Now apply DC supply of 10 V at output port and leave input port open-circuited i.e. $I_1 = 0$ (do the same as in step 1). Measure input and output voltages and currents.



$$V_1 = 6.8281V$$

$$V_2 = 10V$$

$$I_1 = 68.281\mu A$$

$$I_2 = 4.9577mA$$

3. The open-circuit input port refers to the condition for z_{22} and z_{12} . Therefore, using above values and formulae given in Table 12.1, calculate z_{22} and z_{12} .

$$z_{22} = 2.017k \text{ ohms}$$

$$z_{12} = 1.377k \text{ ohms}$$

Express the impedance model in matrix form:

$$\begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} =$$

TASK 1:

$$\begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} = \begin{bmatrix} 1.6308 \text{ k} & 1.377 \text{ k} \\ 1.372 \text{ k} & 2.017 \text{ k} \end{bmatrix}$$

Task 2: To verify impedance –parameters

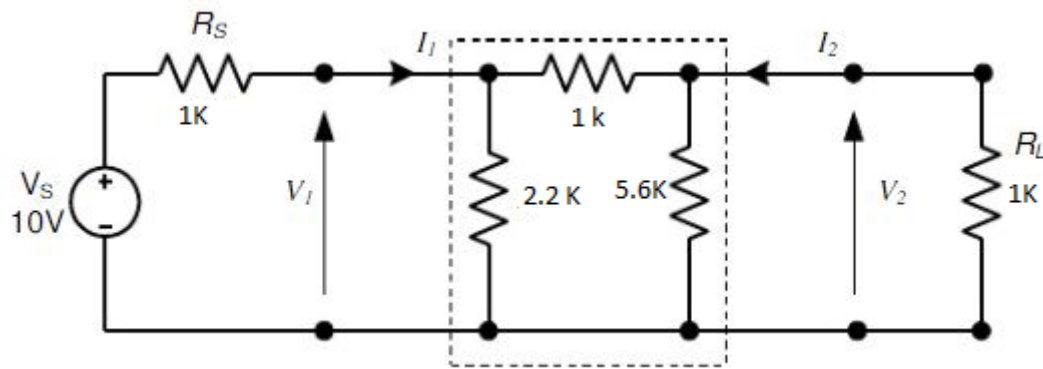
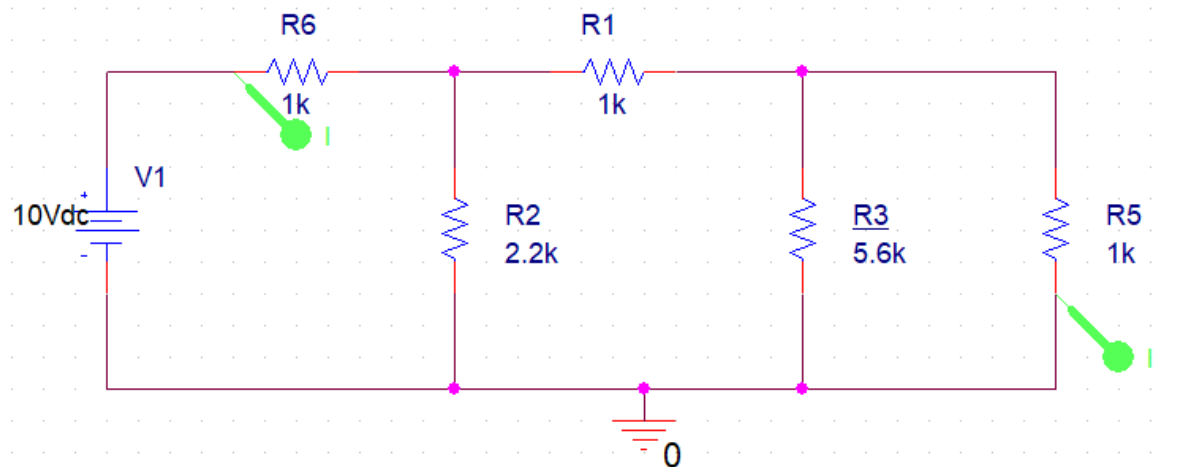
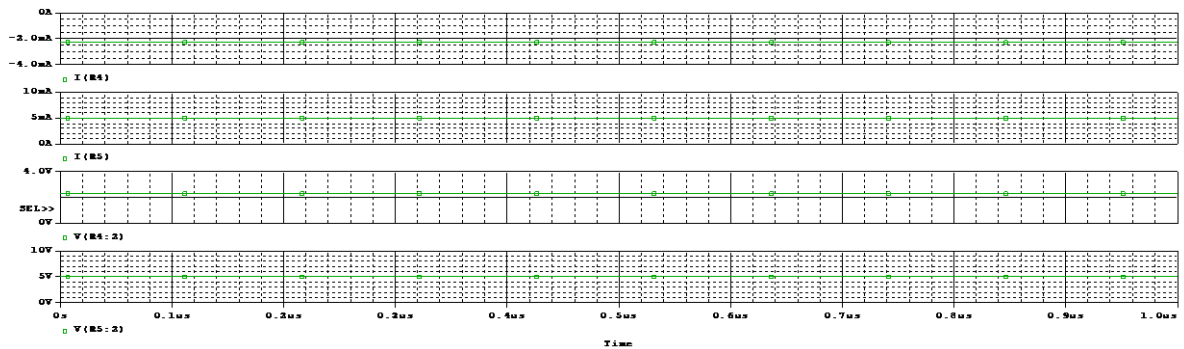


Figure 12.3

- Now connect a load of $1\text{ k}\Omega$ at output port and apply 10 V DC supply at input port and connect $1\text{ k}\Omega$ resistance at input port as shown in Figure 12.3. Measure I_1 and I_2 .





$$I_1 = 4.9888 \text{ mA}$$

$$I_2 = -2.3002 \text{ mA}$$



Now calculate the values of V_1 and V_2 using equations for impedance model and then measure V_1 and V_2 and record in the table given below. **Attach theoretical calculations too.**

TASK 2 :

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 1.6308 \text{ k} & 1.377 \text{ k} \\ 1.372 \text{ k} & 2.017 \text{ k} \end{bmatrix} \begin{bmatrix} 4.9888 \text{ m} \\ -2.3002 \text{ m} \end{bmatrix}$$

$$= \begin{bmatrix} (1.6308 \times 10^3)(4.9888 \times 10^{-3}) + (1.377 \times 10^3)(-2.3002 \times 10^{-3}) \\ (1.372 \times 10^3)(4.9888 \times 10^{-3}) + (2.017 \times 10^3)(-2.3002 \times 10^{-3}) \end{bmatrix}$$

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 4.968 \\ 2.205 \end{bmatrix}$$

Theoretical: $V_1 = 4.968 \text{ V}$, $V_2 = 2.205 \text{ V}$

Table 12.2

Measurement	Theoretical	Experimental
V_1	4.968V	5.0112V
V_2	2.205V	2.3002V

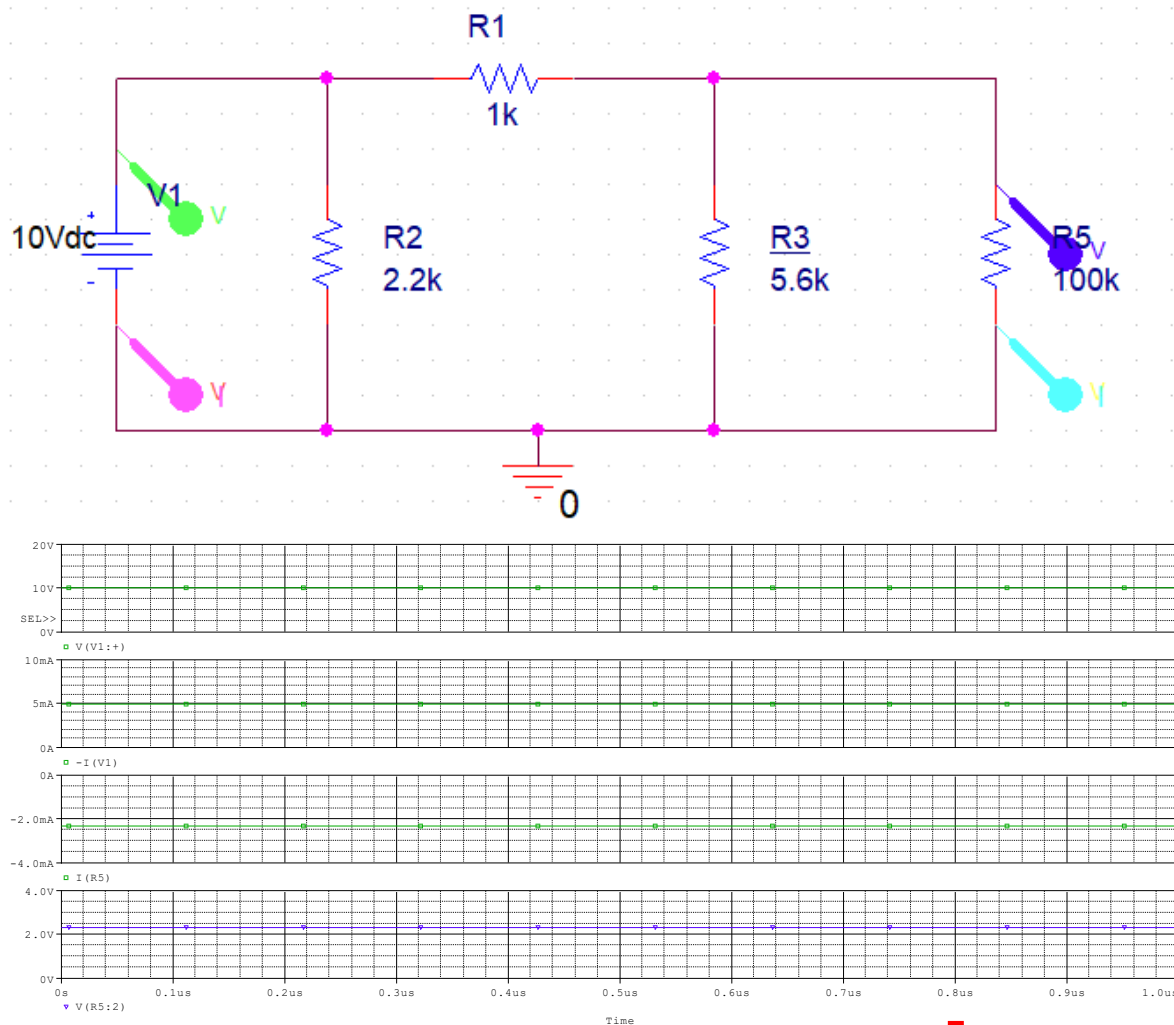


2. Comment on the results.

Theoretical values of V_1 and V_2 obtained through the calculated impedance model align with the experimental values, indicating that the two are equivalent. By utilizing the principles of two-port networks to construct the impedance model, we can conveniently observe how various loads impact the system, without the need for repeated intensive computations, if we possess the essential input and output current data.

Task 3: To find transmission-parameters

1. Apply DC supply of 10 V at input port and leave output port open-circuited i.e. $I_2 = 0$ for the circuit given in Figure 12.2. Now measure input and output voltages and currents.

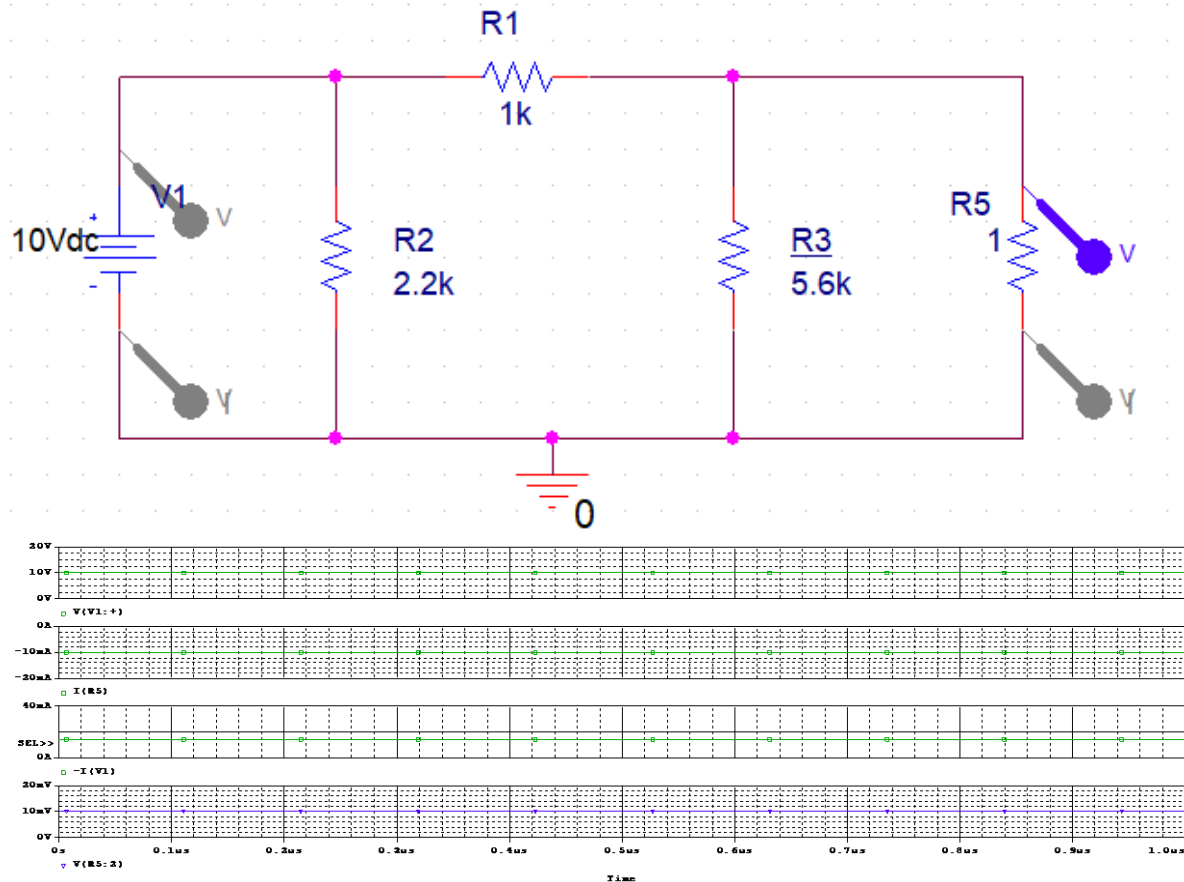


$$V_1 = 10V \quad V_2 = 8.4135V \quad I_1 = 6.132mA \quad I_2 = 84.135\mu A$$

2. The open circuit output refers to the condition for A and C parameters. Therefore, using above values and formulae given in Table 12.1, compute A and C.

$$A = 1.1885 \quad C = 0.7288m \text{ ohms}^{(-1)}$$

3. Now short-circuit the output port i.e., $V_2 = 0$ (place small value of Resistance to make $V_2 = 0$ at output port). Then measure input and output voltages and currents.



$$V_1 = 10V \quad V_2 = 9.988mV$$

$$I_1 = 14.536mA \quad I_2 = -9.988mA$$

4. The short-circuit output port refers to the condition for B and D. Therefore, using above values and formulae given in Table , calculate B and D.

$$B = 1.001k \text{ ohms} \quad D = 1.4553$$

Express the impedance model in matrix form:

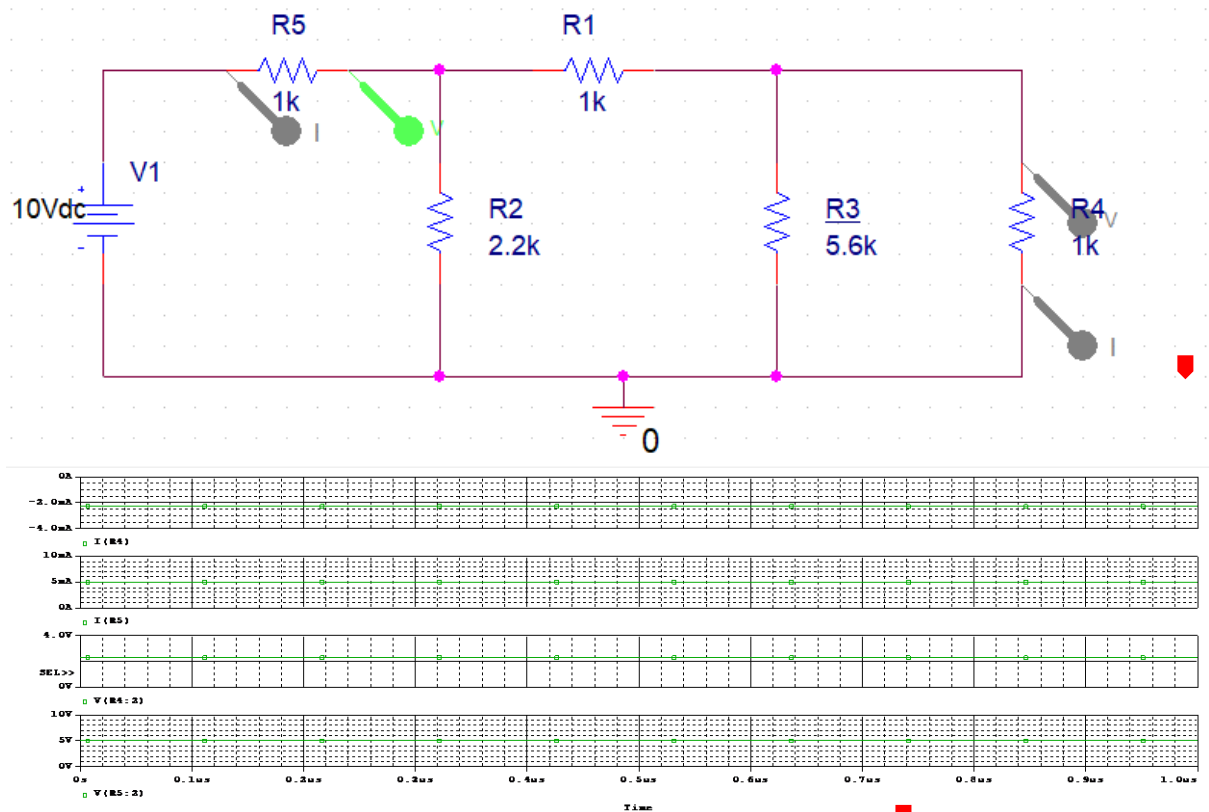
$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} =$$

TASK 3 :

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 1.885 & 1001.2 \\ 0.7288m & 1.4553 \end{bmatrix}$$

Task 4: To verify transmission-parameters

1. Implement the circuit given in Figure 12.3 and measure V_2 and I_2 .



$$V_2 = 2.3002V \quad I_2 = -2.3002mA$$

2. Now calculate the values of V_1 and I_1 using transmission model parameters and record in Table 12.3. **Attach theoretical calculation with the manual.**

TASK 4:

$$V_1 = A V_2 - B I_2$$

$$I_1 = C V_2 - D I_2$$

$$V_1 = (1.885)(2.3002) - (1001.2)(-2.3002 \times 10^{-3})$$

From calculator

$$V_1 = 6.6388 \text{ V}$$

$$I_1 = (0.7288 \times 10^{-3})(2.3002) - (1.4553)(-2.3002 \times 10^{-3})$$

$$I_1 = 5.024 \text{ mA}$$

$$\text{Theoretical} = V_1 = 6.6388 \text{ V}, I_1 = 5.024 \text{ mA}$$

Table 12.3

Measurement	Theoretical	Experimental
V_1	6.6388V	5.0112V
I_1	5.024A	4.988mA

3. Also, record the experimental values of V_1 and I_1 . Comment on the results.

We can deduct from the agreement between our theoretical and experimental values that our transmission model provides a close representation of the real circuit. The transmission model presents advantages, particularly in situations where we only have knowledge of the input response and require an understanding of the output. In contrast, the impedance model demands familiarity with both input and output currents. Additionally, the transmission model can predict responses and determine initial values based on the system's output. Ultimately, the choice of the superior model depends on the available parameters and the parameters of interest to us.

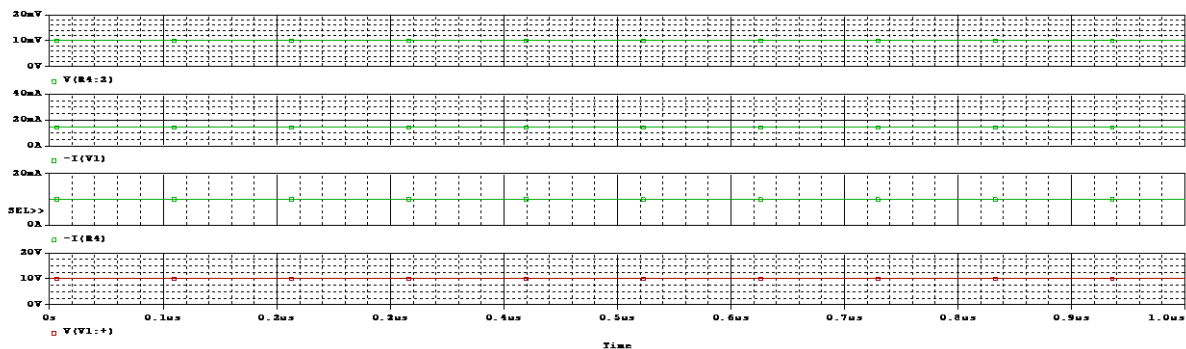
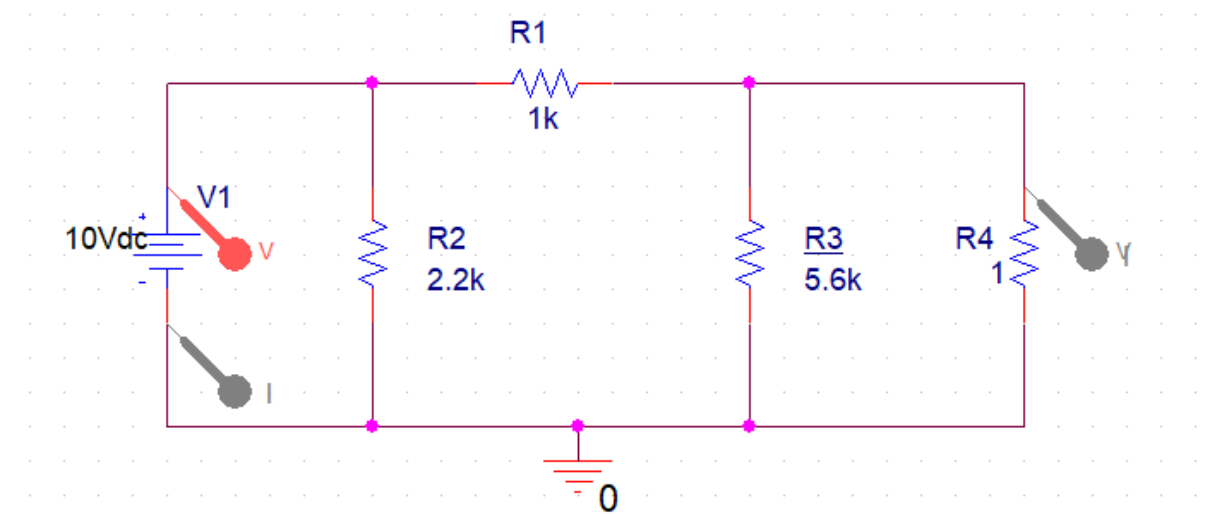
Post-Lab

Task 5: To find admittance model of a two-port network

Simulate the circuit given in in ORCAD PSpice. Determine admittance model parameters i.e., find input and output voltages and currents by short-circuiting input and output ports one by one.

Show all necessary calculations and results. Also attach schematic of the simulated circuit.

When $V_2=0$

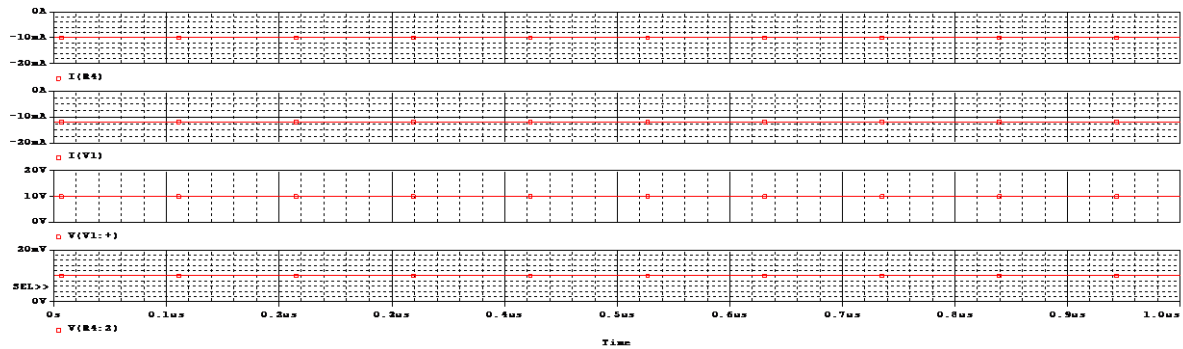
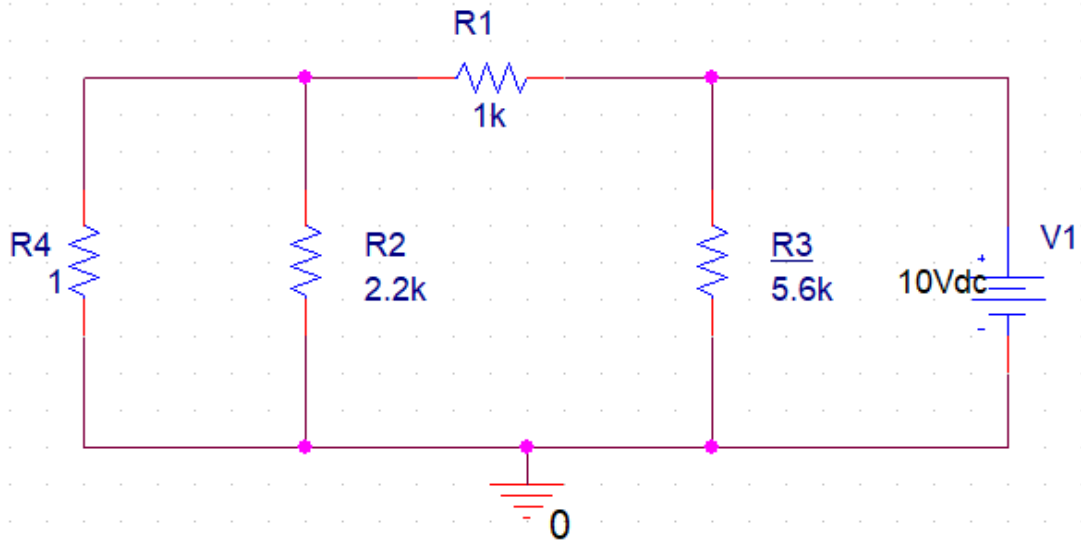


$V_1 = 10V$, $V_2 = 9.988mV$, $I_1 = 14.536mA$, $I_2 = -9.988mA$.

$y_{11} = 1.4536m \text{ ohms}^{(-1)}$, $y_{21} = -0.9988m \text{ ohm}^{(-1)}$.

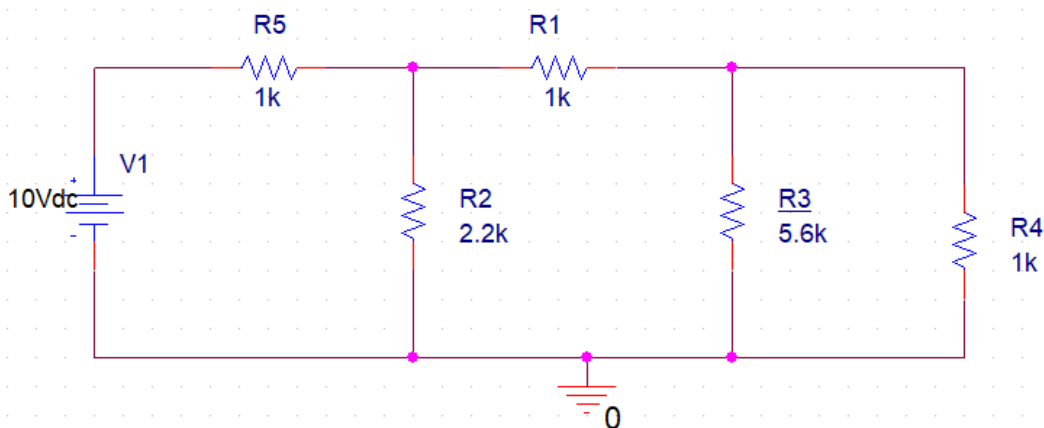
When $V_1=0$:

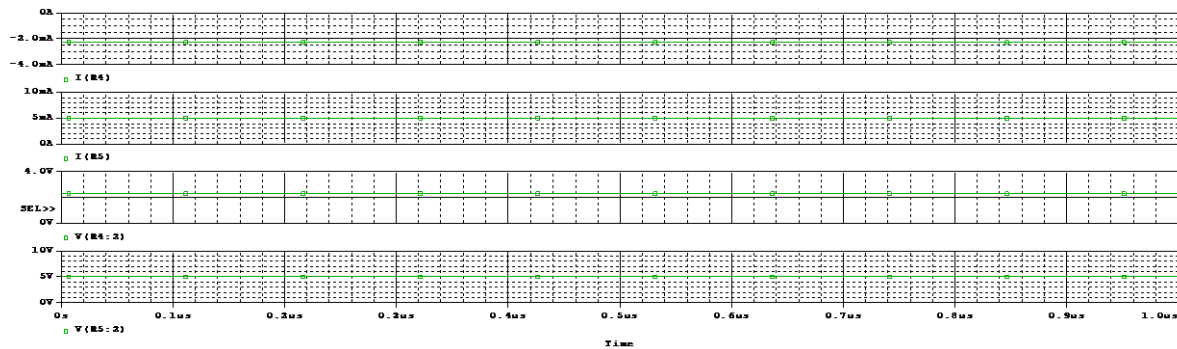




$V1 = 9.98\text{mV}$, $V2 = 10\text{V}$, $I1 = -9.986\text{mA}$, $I2 = 11.776\text{mA}$.

$y12 = -0.9986\text{m ohms}^{(-1)}$, $y22 = 1.1776\text{m ohms}^{(-1)}$





Experimental: $I_1 = 4.998\text{mA}$, $I_2 = -2.3002\text{mA}$, $V_1 = 5.0112\text{V}$, $V_2 = 2.3002\text{V}$.

Theoretical:

$\gamma_{21} = 5.024\text{mA}$

TASK 5:
When $V_2 = 0$:
 $V_1 = 10\text{V}$, $V_2 = 9.988\text{mV}$, $I_1 = 14.536\text{mA}$, $I_2 = -9.988\text{mA}$
 $\gamma_{11} = 1.4536\text{m}\Omega^{-1}$, $\gamma_{21} = -0.9988\text{m}\Omega^{-1}$
When $V_1 = 0$:
 $V_1 = 9.986\text{mV}$, $V_2 = 10\text{V}$, $I_1 = -9.986\text{mA}$, $I_2 = 11.776\text{mA}$
 $\gamma_{12} = -0.9986\text{m}\Omega^{-1}$, $\gamma_{22} = 1.1776\text{m}\Omega^{-1}$

$I_1 = \gamma_{11}V_1 + \gamma_{12}V_2 \rightarrow I_1 = (1.4536 \times 10^{-3})(10) + (-0.9986 \times 10^{-3})(2.3002)$
 $I_1 = (1.4536 \times 10^{-3})(5.0112) + (-0.9986 \times 10^{-3})(2.3002)$
 $I_1 = 4.987\text{mA}$
 $I_2 = \gamma_{21}V_1 + \gamma_{22}V_2$
 $I_2 = (-0.9988 \times 10^{-3})(5.0112) + (1.1776 \times 10^{-3})(2.3002)$
 $I_2 = -2.296\text{mA}$

Theoretical: $I_1 = 4.987\text{mA}$, $I_2 = -2.296\text{mA}$

Measurement	Theoretical	Experimental
I_1	4.987mA	4.988mA
I_2	-2.296mA	-2.3002mA



EE-213 Electric Circuits Lab

Lab 12

Two Port Network

Name: S.M. Asad Ali Student ID: sa09473

Completed
Post-lab
in-lab-
time!

+5

12.1 Objective
To represent a two-port network into its equivalent impedance, transmission and admittance parameter models.

12.2 Introduction
A pair of terminals from any linear electrical network is called port. A single-port network has two terminals i.e., one port. The voltage between the terminals is called port voltage and the current into the terminal is called port current. Similarly, a two-port network has 4 terminals and two ports as represented in Figure 12.1.

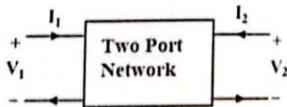


Figure 12.1: Two-Port Network Representation

For analyzing purposes, any electrical network systems such as amplifiers, transistors, filters, matching circuits, transformers, transmission cable etc. can be modeled as a two-port network; input port at left side and output port at right side. A complex circuit when represented by a two-port network becomes a black box. This isolated two-port network simplifies network analysis by representing complex connections and component values in a circuit into their equivalent parameters. To transform a linear circuit into a two-port network, the following conditions must be met.

1. No energy will be stored in two-port network
2. No independent source exists in two-port network
3. The same current must enter and leave a port
4. External connections must be made to either the input or output port i.e., no connections are allowed between the ports

Equivalent Parameters: A two-port network can be modelled using equivalent parameters which are listed below:

1. Impedance z-parameter
2. Admittance y-parameter
3. Hybrid h-parameter

+5 for completing post lab in lab time



Assessment Rubric
Lab 12
Two Port Network

Name:	Student ID:
--------------	--------------------

Points Distribution

Task No.	LR 2 Simulation	LR 4 Data Collection	LR5 Results/Plots	LR 6 Calculation
Task 1	/10	/10	-	/5
Task 2	/5	/5	/5	/5
Task 3	/10	/10	-	/5
Task 4	/5	/5	/5	/5
Task 5(Optional)	-	-	-	-
Total	/30	/30	/10	/20
CLO Mapped	CLO 3	CLO 3	CLO 3	CLO 3

Affective Domain Rubric		Points	CLO Mapped
AR 7	Report Submission	/10	CLO 4

CLO	Total Points	Points Obtained
3	90	
4	10	
Total	100	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.